Two Instruments, One Ecosystem

Open-source bioreactors: what they're *for*, and where OpenEvo goes wrong

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Biopunk Wetware Meeting ▪ `github.com/m9h/biopunk-bioreactor`

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The one-slide thesis

Chi.Bio and OpenEvo look like competitors. They aren't.

- **Chi.Bio is an instrument** — measures a culture richly, in situ, while holding it in a defined state.
- **OpenEvo is an actuator** — pushes a culture through selection, unattended, for weeks.

Both cost ~$300. Both are open. But a machine built for one purpose is *measurably worse* at the other.

The field serves three purposes, not one contest.

Three purposes

P1 – Characterise

Hold a defined state; measure richly in situ; close feedback loops.

Champion: **Chi.Bio**

P2 – Evolve

Weeks of unattended selection; produce *archived, replicated* evidence.

Champion: none yet

P3 – Scale out

Many reliable reactors now; turnkey; supported.

Champions: Pioreactor, eVOLVER

Rich-&-few vs. cheap-&-many | non-contact vs. electrodes-OK | steady-state vs. run-to-stationary

OpenEvo: credit where due

A real contribution — these are genuine strengths:

- Peak-to-peak growth readout tied to hardware dilution events (no local-maxima noise).
- 940 nm OD avoids optogenetic & ambient overlap; $R^2 \geq 0.98$ to $\mathrm{OD}_{600} = 6$.
- Closed autoclavable vessel — better sterility than Chi.Bio's open tube.
- 4,000-hour remote run; controlled from another continent.
- Assembly manual validated by an undergrad with no electronics experience.

The cheapest buildable three-media selection cycler in the field.
The problems below are correctable — framing and analysis, not hardware.

OpenEvo: two verifiable errors

Error 1 — the headline figure contradicts itself

Abstract: "55% faster." Results: "4.9 vs 7.6 h, a 36% increase in growth rate."
Same data. $7.6 \to 4.9\,\text{h}$ is a *36% shorter doubling time* = a *55% higher rate* ($1/4.9 \div 1/7.6 = 1.55$). The results sentence mislabels one as the other.

Error 2 — a false claim about a competitor

"Pioreactor. . . only the *software* is open source."
False: the hardware is openly licensed (CC BY-SA 4.0, per its `LICENSE`).
(Our own first draft misnamed it CERN-OHL — a contributor corrected it against the primary file.)

OpenEvo: the deeper gap — evidence

$n = 1$

One clone, one population sequenced. No replicates $\Rightarrow$ no convergence $\Rightarrow$ *cannot say which mutation caused the adaptation.* The LTEE runs **twelve** for this reason.

No archiving

One hand-taken glycerol stock. The LTEE's real innovation was the *freezer*, not the flask — revive an ancestor, race it against a descendant. Without it, fitness is unmeasurable.

OpenEvo cites Barrick's "…Archiving Populations, and Measuring Fitness" — then implements neither.

The matrix (abridged)

	Chi.Bio	OpenEvo	Pioreactor	eVOLVER	EVE
Purpose	P1	P2	P3	P3	P2
Paper	yes	yes	none	yes	yes
Parallelism	8	1	cluster	16	3+
Rich sensing	spectro.	OD	plugin	config.	OD
Archiving	no	no	no	no	no
Cost	\$300	\$300	\$329	<\$2k/16	\$115
Build	PCB	guided	buy	shop	mod.

Two patterns: nobody archives (the field's blind spot) · the only paperless platform is the only pure product.

The proposal: build two, share the parts

Characterisation machine

Non-contact is inviolable.

- + optode DO / pH (read through wall)
- + dispersive spectrometer 340–850 nm
- + sealed gas-controlled lid

Evolution machine

Cheap × many; resist sensing bloat.

- + 8 replicate chambers, one interface
- + **automated chilled archive** (nobody has built this)
- + batch-excursion mode; `breseq`

Why two? The sensors sort themselves by contact.

Optical (DO/pH/spectrometry) → characterisation. Electrode (conductivity/ERT/viable-cell) → evolution. *Share* the module footprint, sensor designs, firmware, calibration — not the vessel.

These are teaching instruments too

- **EVE** — eLife paper is literally "...and its educational use"; <$200; validated in a *real classroom* evolving *E. coli* under chloramphenicol.
- **Pioreactor** — educators portal; one device teaches micro + EE + CS at once.
- **OpenEvo** — the manual *is* a course; anchored a global hackathon.
- **Chi.Bio** — "remove the lid, observe fluorescence by eye."

But the teaching inherits the research blind spots — archiving, replicates, and carrying capacity are as under-taught as they are under-built.

Take-aways / call to build

1. Stop comparing these as one contest — **three purposes**, different winners.
2. Evolution has no champion: nobody archives or replicates. *That's the opening.*
3. The killer feature nobody has shipped: an **automated chilled fraction collector**.
4. OpenEvo's errors are real but fixable — and we filed them, fairly, with fixes.

Public, CC BY 4.0, corrections welcome — a PR already landed and fixed one of *my* errors.

github.com/m9h/biopunk-bioreactor